Leibniz Integral Rule

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0 The Leibniz Integral Rule

Theorem 0.1 (Leibniz Integral Rule – Riemann Integral)**.** Suppose $I \subseteq \mathbb{R}$ is an open interval and $f : [a, b] \times I \to \mathbb{R}$, $(x, t) \mapsto f(x, t)$ such that f and $\frac{\partial f}{\partial t}$ are continuous on $[a, b] \times I$. Define $S : I \to \mathbb{R}$ as follows:

$$S(t) = \int_a^b f(x, t)\, dx$$

Then S is differentiable on I and:

$$S'(t) = \int_a^b \frac{\partial f}{\partial t}(x, t)\, dx$$

Proof. Suppose $t_0 \in I$ and $\varepsilon > 0$. Since I is an open interval, there exists $\delta_0 > 0$ such that $[t_0 - \delta_0, t_0 + \delta_0] \subseteq I$. Since $\frac{\partial f}{\partial t}$ is continuous on $[a, b] \times I$, by the Heine-Cantor theorem, it is uniformly continuous on $[a, b] \times [t_0 - \delta_0, t_0 + \delta_0]$. Thus there exists $0 < \delta < \delta_0$ such that:

$$\left| \frac{\partial f}{\partial t}(x, s) - \frac{\partial f}{\partial t}(x, t_0) \right| < \frac{\varepsilon}{b - a} \text{ for all } x \in [a, b] \text{ and all } s \in I, |s - t_0| < \delta$$

Now suppose $t \in I$, $|t - t_0| < \delta$ and $x \in [a, b]$. By Lagrange's mean value theorem, there exists ξ (possibly depending on x) between t_0 and t such that $\frac{f(x,t) - f(x,t_0)}{t - t_0} = \frac{\partial f}{\partial t}(x, \xi)$. This yields:

$$\left| \frac{f(x, t) - f(x, t_0)}{t - t_0} - \frac{\partial f}{\partial t}(x, t_0) \right| = \left| \frac{\partial f}{\partial t}(x, \xi) - \frac{\partial f}{\partial t}(x, t_0) \right| < \frac{\varepsilon}{b - a}$$

Note that the above inequality holds for all $x \in [a, b]$, even though ξ may depend on x. We also have:

$$\begin{aligned} \frac{S(t) - S(t_0)}{t - t_0} - \int_a^b \frac{\partial f}{\partial t}(x, t_0)\, dx &= \frac{1}{t - t_0} \left(\int_a^b f(x, t)\, dx - \int_a^b f(x, t_0)\, dx \right) - \int_a^b \frac{\partial f}{\partial t}(x, t_0)\, dx \\ &= \int_a^b \left(\frac{f(x, t) - f(x, t_0)}{t - t_0} - \frac{\partial f}{\partial t}(x, t_0) \right) dx \end{aligned}$$

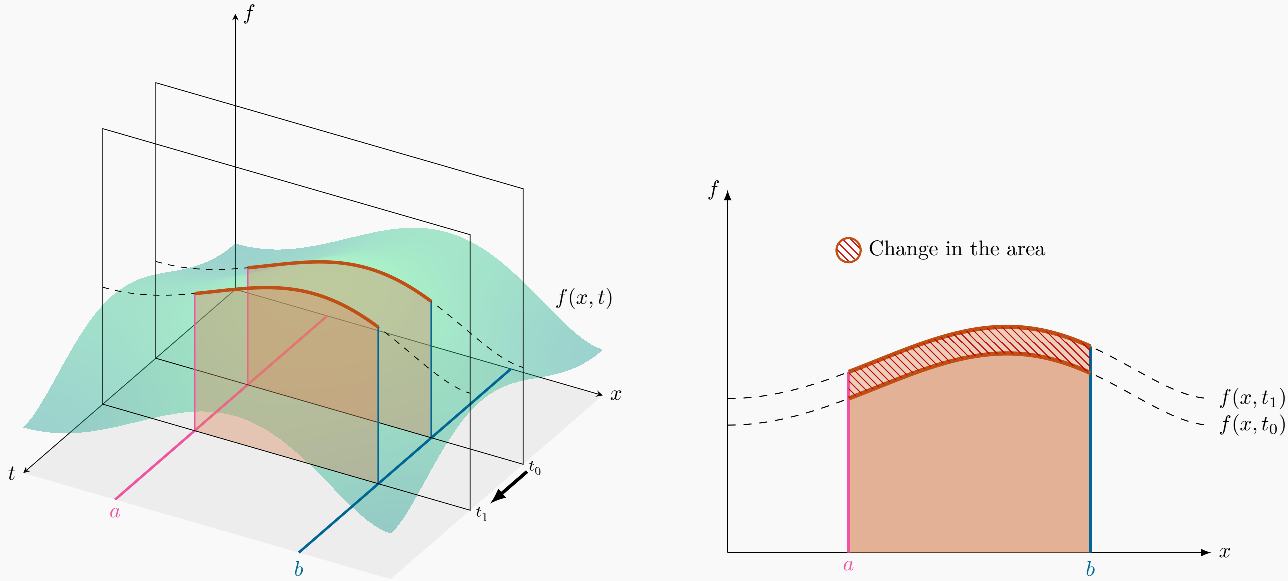
This yields:

$$\left| \frac{S(t) - S(t_0)}{t - t_0} - \int_a^b \frac{\partial f}{\partial t}(x, t_0)\, dx \right| \le \int_a^b \left| \frac{f(x, t) - f(x, t_0)}{t - t_0} - \frac{\partial f}{\partial t}(x, t_0) \right| dx < \int_a^b \frac{\varepsilon}{b - a}\, dx = (b - a)\frac{\varepsilon}{b - a} = \varepsilon$$

Thus S is differentiable at t_0 and $S'(t_0) = \int_a^b \frac{\partial f}{\partial t}(x, t_0)\, dx$. 😎

The Leibniz Integral Rule

$$\boxed{\frac{d}{dt}\left(\int_a^b f(x,t)\,dx\right) = \int_a^b \frac{\partial f}{\partial t}(x,t)\,dx}$$



As t moves from t_0 to t_1, the shaded area changes from $\displaystyle\int_a^b f(x,t_0)\,dx$ to $\displaystyle\int_a^b f(x,t_1)\,dx$.

Dividing this by $t_1 - t_0$ and taking the limit as $t_1 - t_0$ becomes small, we get $\displaystyle\int_a^b \frac{\partial f}{\partial t}(x,t)\,dx$.

Theorem 0.2 (Leibniz Integral Rule – Improper Riemann Integral)**.** Suppose $I \subseteq \mathbb{R}$ is an open interval and $f : [a,\infty)\times I \to \mathbb{R}$, $(x,t) \mapsto f(x,t)$ such that:

- f is continuous on $[a,\infty)\times I$ improperly Riemann integrable with respect to x on $[a,\infty)$ for each $t \in I$.
- $\frac{\partial f}{\partial t}$ is continuous on $[a,\infty)\times I$ and improperly Riemann integrable with respect to x on $[a,\infty)$, uniformly in $t \in I$. In other words, for any $\varepsilon > 0$, there exists $M > a$ (independent of t) such that for all $y > M$ and all $t \in I$, we have $\left|\int_a^y f(x,t)\,dx\right| < \varepsilon$.

Define $S : I \to \mathbb{R}$ as follows:

$$S(t) = \int_a^\infty f(x,t)\,dx$$

Then S is differentiable on I and:

$$S'(t) = \int_a^\infty \frac{\partial f}{\partial t}(x,t)\,dx$$

Proof. Suppose $(b_n)_{n=1}^\infty$ is a sequence in (a,∞) and $\lim_{n\to\infty} b_n = \infty$. For each $n \in \mathbb{N}$, define $S_n : I \to \mathbb{R}$ as follows:

$$S_n(t) = \int_a^{b_n} f(x,t)\,dt$$

By **Theorem 0.1**, for each $n \in \mathbb{N}$, we have $S_n'(t) = \int_a^{b_n} \frac{\partial f}{\partial t}(x,t)\,dt$. Since $\frac{\partial f}{\partial t}(x,t)$ is improperly Riemann integrable with respect to x on $[a,\infty)$, uniformly in t, we have:

$$\lim_{n\to\infty} S_n'(t) = \int_a^\infty \frac{\partial f}{\partial t}(x,t)\,dt$$

FINISH THE PROOF!!! 😎

The Leibniz integral rule can be applied to known integrals to derive related formulas:

Example.

$$
\begin{aligned}
\int_0^\infty \frac{1}{x^2+a^2}\,dx = \frac{\pi}{2a} \quad &\xrightarrow{\frac{d}{da}} \quad \int_0^\infty \frac{1}{(x^2+a^2)^2}\,dx = \frac{\pi}{4a^3} \\
&\xrightarrow{\frac{d}{da}} \quad \int_0^\infty \frac{1}{(x^2+a^2)^3}\,dx = \frac{3\pi}{16a^5} \\
&\vdots \\
&\xrightarrow{\frac{d}{da}} \quad \int_0^\infty \frac{1}{(x^2+a^2)^n}\,dx = \binom{2n-2}{n-1}\frac{\pi}{(2a)^{2n-1}}
\end{aligned}
$$

Example.

$$
\begin{aligned}
\int_0^\infty e^{-ax}\,dx = \frac{1}{a} \quad &\xrightarrow{\frac{d}{da}} \quad \int_0^\infty xe^{-ax}\,dx = \frac{1}{a^2} \\
&\xrightarrow{\frac{d}{da}} \quad \int_0^\infty x^2e^{-ax}\,dx = \frac{2}{a^3} \\
&\vdots \\
&\xrightarrow{\frac{d}{da}} \quad \int_0^\infty x^ne^{-ax}\,dx = \frac{n!}{a^{n+1}}
\end{aligned}
$$

The Leibniz integral rule can also be used to solve certain integrals, by differentiating them first:

Example.

$$S(t) = \int_0^1 \frac{x^t-1}{\log(x)}\,dx \qquad (t > -1)$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^1 x^t\,dx = \frac{1}{t+1}$$

Integrating yields:

$$S(t) = \log(t+1) + C$$

Where C is a constant. Note that the integrand is zero when $t = 0$, so $S(0) = 0$ and so $C = 0$. Thus:

$$\int_0^1 \frac{x^t-1}{\log(x)}\,dx = \log(t+1)$$

Example (Dirichlet integral)**.**

$$S(t) = \int_0^\infty e^{-tx}\frac{\sin(x)}{x}\,dx \qquad (t > 0)$$

Differentiating under the integral, we get:

$$S'(t) = -\int_0^\infty e^{-tx}\sin(x)\,dx = -\frac{1}{t^2+1}$$

Integrating yields:

$$S(t) = -\arctan(t) + C$$

Where C is a constant. Note that the integrand tends to zero as $t \to \infty$, so $\lim_{t\to\infty} S(t) = 0$ and so $C = \frac{\pi}{2}$. Thus:

$$\int_0^\infty e^{-tx}\frac{\sin(x)}{x}\,dx = \frac{\pi}{2} - \arctan(t)$$

Taking the limit as $t \to 0^+$ yields:

$$\int_0^\infty \frac{\sin(x)}{x}\,dx = \frac{\pi}{2}$$

The following integral was solved in 1844 by Joseph Serret (1819 – 1885).

Example (Serret's Integral)**.** Consider the following integral:

$$I = \int_0^1 \frac{\log(1+x)}{1+x^2}\,dx$$

To evaluate this, we first define $S(t)$ as follows:

$$S(t) = \int_0^1 \frac{\log(1+tx)}{1+x^2}\,dx$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^1 \frac{x}{(1+tx)(1+x^2)}\,dx = -\frac{\log(1+t)}{1+t^2} + \frac{\pi t}{4(1+t^2)} + \frac{\log(2)}{2(1+t^2)}$$

Note that $S(0) = 0$ and $S(1) = I$. Thus, we have:

$$\begin{aligned} I = S(1) - S(0) &= \int_0^1 S'(t)\,dt \\ &= -\int_0^1 \frac{\log(1+t)}{1+t^2}\,dt + \frac{\pi}{4}\int_0^1 \frac{t}{1+t^2}\,dt + \frac{\log(2)}{2}\int_0^1 \frac{1}{1+t^2}\,dt \\ &= -\int_0^1 \frac{\log(1+t)}{1+t^2}\,dt + \frac{\pi}{4}\left(\frac{\log(2)}{2}\right) + \frac{\log(2)}{2}\left(\frac{\pi}{4}\right) \\ &= -I + \frac{\pi}{4}\log(2) \end{aligned}$$

Rearranging yields $2I = \frac{\pi}{4}\log(2)$. Thus:

$$\int_0^1 \frac{\log(1+x)}{1+x^2}\,dx = \frac{\pi}{8}\log(2)$$

Example.

$$S(t) = \int_0^1 \frac{\log(1+tx)}{x\sqrt{1-x^2}}\,dx \qquad (t \geq -1)$$

Differentiating under the integral (and assuming $-1 \leq t \leq 1$ for now), we get:

$$S'(t) = \int_0^1 \frac{1}{(1+tx)\sqrt{1-x^2}}\,dx = \frac{\arccos(t)}{\sqrt{1-t^2}}$$

Integrating yields:

$$S(t) = -\frac{\arccos(t)^2}{2} + C$$

Where C is a constant. Note that the integrand is zero when $t = 0$, so $S(0) = 0$ and so $C = \frac{\pi^2}{8}$. Thus:

$$\int_0^1 \frac{\log(1+tx)}{x\sqrt{1-x^2}}\,dx = \frac{\pi^2}{8} - \frac{\arccos(t)^2}{2}$$

If $t > 1$, we instead have:

$$S'(t) = \frac{\operatorname{arccosh}(t)}{\sqrt{t^2-1}}$$

This time, integrating yields:

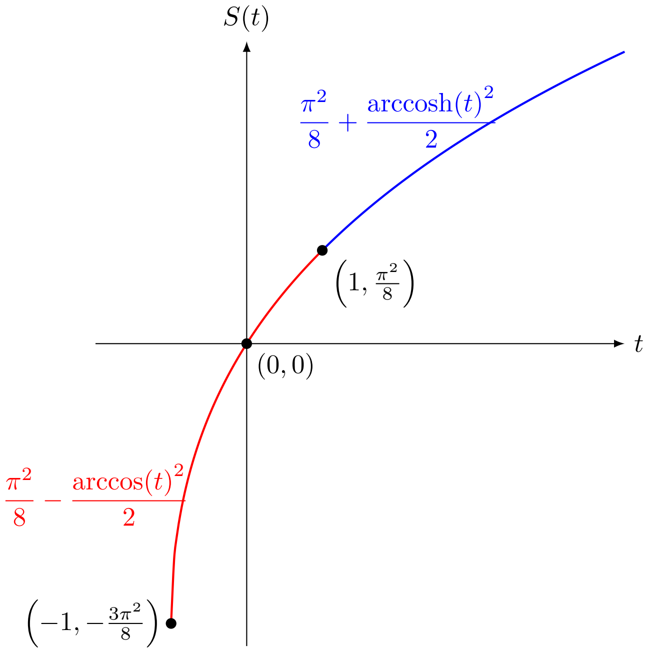
$$S(t) = \frac{\operatorname{arccosh}(t)^2}{2} + K$$

Where K is a constant. As we have already shown above, we have $S(1) = \frac{\pi^2}{8}$ and so $K = \frac{\pi^2}{8}$. Thus:

$$\int_0^1 \frac{\log(1+tx)}{x\sqrt{1-x^2}}\,dx = \begin{cases} \frac{\pi^2}{8} - \frac{\arccos(t)^2}{2} & -1 \leq t \leq 1 \\ \frac{\pi^2}{8} + \frac{\operatorname{arccosh}(t)^2}{2} & t > 1 \end{cases}$$

Note that the function on the right, although defined piecewise, is analytic at every $t > -1$ (including $t = 1$).

$$\sum_{n=1}^{\infty} \frac{(-2)^{n+1}((n-1)!)^2}{(2n)!}(x-1)^n = \begin{cases} -\arccos(x)^2 & -1 \leq x \leq 1 \\ \operatorname{arccosh}(x)^2 & 1 \leq x \leq 3 \end{cases}$$



Example.

$$S(t) = \int_0^{\frac{\pi}{2}} \frac{\arctan(t\tan(x))}{\tan(x)}\,dx$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^{\frac{\pi}{2}} \frac{1}{1+t^2\tan(x)^2}\,dx = \frac{\pi}{2(t+1)}$$

Integrating yields:

$$S(t) = \frac{\pi}{2}\log(t+1) + C$$

Where C is a constant. Note that the integrand is zero when $t=0$, so $S(0)=0$ and so $C=0$. Thus:

$$\int_0^{\frac{\pi}{2}} \frac{\arctan(t\tan(x))}{\tan(x)}\,dx = \frac{\pi}{2}\log(t+1)$$

In particular, setting $t=1$ yields:

$$\int_0^{\frac{\pi}{2}} \frac{x}{\tan(x)}\,dx = \frac{\pi}{2}\log(2)$$

Example.

$$S(t) = \int_0^{\infty} e^{-x^2}\cos(tx)\,dx$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^{\infty} -xe^{-x^2}\sin(tx)\,dx$$

Integrating by parts yields:

$$S'(t) = \left[-\frac{1}{2}e^{-x^2}\sin(tx)\right]_{x=0}^{x=\infty} - \int_0^{\infty} -\frac{t}{2}e^{-x^2}\cos(tx)\,dx = \frac{t}{2}\int_0^{\infty} e^{-x^2}\cos(tx)\,dx$$

Thus $S'(t) = \frac{t}{2}S(t)$. This is a linear first-order ODE in $S(t)$, whose solution is $S(t) = Ce^{-\frac{t^2}{4}}$. To find C, note that:

$$S(0) = \int_0^{\infty} e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}$$

Thus:

$$\int_0^{\infty} e^{-x^2}\cos(tx)\,dx = \frac{\sqrt{\pi}}{2}e^{-\frac{t^2}{4}}$$

Example. This integral was discovered by Laplace in 1810:

$$S(t) = \int_0^{\infty} \frac{\cos(tx)}{x^2+a^2}\,dx$$

Differentiating under the integral, we get:

$$S'(t) = \int_0^\infty -\frac{x\sin(tx)}{x^2+a^2}\,dx \qquad \text{(Convergent for } t \neq 0\text{)}$$

Differentiating again, we get:

$$S''(t) = \int_0^\infty -\frac{x^2\cos(tx)}{x^2+a^2}\,dx \qquad \text{(Divergent)}$$

This is a problem, as the integral on the right does not converge. Instead, we can integrate the expression for $S(t)$ by parts:

$$S(t) = \left[\frac{\sin(tx)}{t(x^2+a^2)}\right]_{x=0}^{x=\infty} - \int_0^\infty -\frac{2x\sin(tx)}{t(x^2+a^2)^2}\,dx = \frac{2}{t}\int_0^\infty \frac{x\sin(tx)}{(x^2+a^2)^2}\,dx$$

Call the integral on the right $K(t)$. Now we can differentiate under the integral:

$$\begin{aligned} K'(t) &= \int_0^\infty \frac{x^2\cos(tx)}{(x^2+a^2)^2}\,dx \\ &= \int_0^\infty \frac{\cos(tx)}{x^2+a^2}\,dx - a^2\int_0^\infty \frac{\cos(tx)}{(x^2+a^2)^2}\,dx \\ &= S(t) - a^2\int_0^\infty \frac{\cos(tx)}{(x^2+a^2)^2}\,dx \end{aligned}$$

Since $K(t) = \frac{t}{2}S(t)$, we can rewrite the above equation as follows:

$$\begin{aligned} \frac{t}{2}S'(t) + \frac{1}{2}S(t) &= S(t) - a^2\int_0^\infty \frac{\cos(tx)}{(x^2+a^2)^2}\,dx \\ \frac{t}{2}S'(t) - \frac{1}{2}S(t) &= -a^2\int_0^\infty \frac{\cos(tx)}{(x^2+a^2)^2}\,dx \end{aligned}$$

Differentiating under the integral again, we get:

$$\frac{t}{2}S''(t) + \frac{1}{2}S'(t) - \frac{1}{2}S'(t) = a^2\int_0^\infty \frac{x\sin(tx)}{(x^2+a^2)^2}\,dx$$

The integral on the right is simply $K(t) = \frac{t}{2}S(t)$. This yields:

$$\frac{t}{2}S''(t) = a^2\frac{t}{2}S(t)$$

Thus $S''(t) = a^2S(t)$. This is a linear second-order ODE in $S(t)$ with constant coefficients, whose solution is $S(t) = C_1e^{at} + C_2e^{-at}$. To find C_1 and C_2, we need two equations. First, we have:

$$S(0) = \int_0^\infty \frac{1}{x^2+a^2}\,dx = \frac{\pi}{2a}$$

Thus $C_1 + C_2 = \frac{\pi}{2a}$. Second, we have:

$$\lim_{t\to\infty} S(t) = \lim_{t\to\infty}\frac{2}{t}\int_0^\infty \frac{x\sin(tx)}{(x^2+a^2)^2}\,dx = 0$$

This can also be seen by applying the Riemann-Lebesgue lemma directly to $S(t)$. Thus $C_1 = 0$, and so $C_2 = \frac{\pi}{2a}$. This yields:

$$S(t) = \frac{\pi}{2a}e^{-at} \qquad (t \geq 0)$$

However, this is only valid for $t \geq 0$, as $S(t)$ is not differentiable at $t = 0$. To handle the case $t < 0$, we can use the fact that $S(t)$ is an even function of t (or alternatively, use the values $S(0) = \frac{\pi}{2a}$ and $\lim_{t\to-\infty} S(t) = 0$ to solve for C_1 and C_2). Thus, we have:

$$S(t) = \frac{\pi}{2a}e^{-a|t|}$$

Example. Define $f : [0,1]\times\mathbb{R}\to\mathbb{R}$ as follows:

$$f(x,t) = \begin{cases} \dfrac{xt^3}{\left(x^2+t^2\right)^2} & (x,t)\neq(0,0) \\ 0 & (x,t) = (0,0) \end{cases}$$

Then for all $t\in\mathbb{R}$, we have $S(t) = \int_0^1 f(x,t)\,dx = \frac{t}{2(t^2+1)}$. Thus $S'(t) = \frac{1-t^2}{2(t^2+1)^2}$, and so $S'(0) = \frac{1}{2}$. However,

$$\frac{\partial f}{\partial t} = \begin{cases} \dfrac{xt^2\left(3x^2-t^2\right)}{\left(x^2+t^2\right)^3} & (x,t)\neq(0,0) \\ 0 & (x,t) = (0,0) \end{cases} \qquad \therefore \quad \int_0^1 \frac{\partial f}{\partial t}(x,0)\,dx = \int_0^1 0\,dx = 0$$

The problem here is that $\frac{\partial f}{\partial t}$ is not continuous (as a function of two variables) at $(0,0)$, as $\frac{\partial f}{\partial t}(t,t) = \frac{1}{4t}$ (for $t\neq 0$), which blows up as $t\to 0^+$.

1 Other versions

The Leibniz integral rule can be reformulated using the Lebesgue integral:

Theorem 1.1 (Leibniz Integral Rule – Lebesgue Integral). Suppose $(X, \mathcal{A}, \mu)$ is a measure space, $I \subseteq \mathbb{R}$ is an open interval and $f : X \times I \to \mathbb{R}$ satisfies the following:

1. The function $x \mapsto f(x,t)$ is integrable for all $t \in I$
2. The function $f \mapsto f(x,t)$ is differentiable for all $x \in X$
3. There is an integrable function $g : X \to \mathbb{R}$ such that $\left|\frac{\partial f}{\partial t}(x,t)\right| \leq g(x)$ for all $x \in X$ and all $t \in I$

Then the function $x \mapsto \frac{\partial f}{\partial t}$ is integrable for all $t \in U$, and:

$$\frac{d}{dt}\left(\int_X f(x,t)\,d\mu(x)\right) = \int_X \frac{\partial f}{\partial t}(x,t)\,d\mu(x)$$

Proof. Suppose $(h_n)_{n=1}^{\infty}$ is a sequence in $\mathbb{R}$ and $\lim_{n\to\infty} h_n = 0$. For each $n \in \mathbb{N}$, define $g_n : X \to \mathbb{R}$ as follows:

$$g_n(x) = \frac{f(x,t+h_n) - f(x,t)}{h_n} - \frac{\partial f}{\partial t}(x,t)$$

Then $\lim_{n\to\infty} g_n(x) = 0$ for all $x \in X$. By Lagrange's mean value theorem, we have:

$$\frac{f(x,t+h_n) - f(x,t)}{h_n} = \frac{\partial f}{\partial t} \text{ for some } \tau_n \text{ between } t \text{ and } t + h_n$$

Thus for all $n \in \mathbb{N}$ and all $x \in X$, we have:

$$|g_n(x)| = \left|\frac{\partial f}{\partial t} - \frac{\partial f}{\partial t}(x,t)\right| \leq \left|\frac{\partial f}{\partial t}\right| + \frac{\partial f}{\partial t}(x,t) \leq g(x) + g(x) = 2g(x)$$

By the dominated convergence theorem, we have:

$$\frac{1}{h_n}\int_X (f(x,t+h_n) - f(x,t))\,d\mu(x) - \int_X \frac{\partial f}{\partial t}(x,t)\,d\mu(x) = \int_X g_n(x)\,d\mu(x) \to 0 \text{ as } n \to \infty$$

As $n \to \infty$, the first integral tends to $\frac{d}{dt}\left(\int_X f(x,t)\,d\mu(x)\right)$. Since this holds for every sequence $(h_n)_{n=1}^{\infty}$ that converges to zero, we have:

$$\frac{d}{dt}\left(\int_X f(x,t)\,d\mu(x)\right) = \int_X \frac{\partial f}{\partial t}(x,t)\,d\mu(x)$$

😎

The Leibniz integral rule can also be adapted to the case where the limits are not constants but functions of t:

Theorem 1.2 (Leibniz Integral Rule – Variable Limits). Suppose $I \subseteq \mathbb{R}$ is an open interval, $a, b : I \to \mathbb{R}$ are differentiable on I such that $a(t), b(t) \in [A,B]$ for all $t \in I$, and $f : [a,b] \times I \to \mathbb{R}$, $(x,t) \mapsto f(x,t)$ such that f and $\frac{\partial f}{\partial t}$ are continuous on $[A,B] \times I$. Define $S : I \to \mathbb{R}$ as follows:

$$S(t) = \int_{a(t)}^{b(t)} f(x,t)\,dx$$

Then S is differentiable on I and:

$$S'(t) = b'(t)f(b(t),t) - a'(t)f(a(t),t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x,t)\,dx$$

Proof. Define $G : [A,B] \times [A,B] \times I \to \mathbb{R}$ as follows:

$$G(a,b,t) = \int_a^b f(x,t)\,dx$$

In other words, G is a function of three independent variables a, b, t. By the fundamental theorem of calculus, we have:

$$\frac{\partial G(a,b,t)}{\partial b} = f(b,t) \qquad \frac{\partial G(a,b,t)}{\partial a} = -f(a,t)$$

By **Theorem 0.1**, we also have:

$$\frac{\partial G(a,b,t)}{\partial t} = \int_a^b \frac{\partial f}{\partial t}(x,t)\,dx$$

Thus, by the multivariable chain rule, we have:

$$\begin{aligned}\frac{d}{dt}G(a(t),b(t),t) &= \frac{db(t)}{dt}\frac{\partial G}{\partial b}(a(t),b(t),t) + \frac{da(t)}{dt}\frac{\partial G}{\partial a}(a(t),b(t),t) + \frac{dt}{dt}\frac{\partial G(a,b,t)}{\partial t} \\ &= b'(t)\frac{\partial G}{\partial b}(a(t),b(t),t) + a'(t)\frac{\partial G}{\partial a}(a(t),b(t),t) + \frac{\partial G}{\partial t}(a(t),b(t),t) \\ &= b'(t)f(b(t),t) - a'(t)f(a(t),t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x,t)\,dx\end{aligned}$$

😎

Fix $t_0 \in I$.

$$\begin{aligned}S(t) &= \int_{a(t)}^{b(t)} f(x,t)\,dx \\ &= \int_{b(t_0)}^{b(t)} f(x,t)\,dx - \int_{a(t_0)}^{a(t)} f(x,t)\,dx + \int_{a(t_0)}^{b(t_0)} f(x,t)\,dx \\ &= \int_{b(t_0)}^{b(t)} f(x,t_0)\,dx - \int_{a(t_0)}^{a(t)} f(x,t_0)\,dx + \int_{a(t_0)}^{b(t_0)} f(x,t)\,dx \\ &\quad + \int_{b(t_0)}^{b(t)} (f(x,t)-f(x,t_0))\,dx - \int_{a(t_0)}^{a(t)} (f(x,t)-f(x,t_0))\,dx\end{aligned}$$

We first show that $S_1'(t) = b'(t)f(b(t),t)$. By the fundamental theorem of calculus and the chain rule, we have:

$$S_1'(t) = b'(t)f(b(t),t) \qquad\qquad S_2'(t) = a'(t)f(a(t),t)$$

By **Theorem 0.1**, we also have:

$$S_3'(t) = \int_{a(t_0)}^{b(t_0)} \frac{\partial f}{\partial t}(x,t)\,dx$$

Since b is differentiable at t_0, it is continuous at t_0, so $\lim_{t\to t_0} |b(t)-b(t_0)| = 0$. Since $\frac{\partial f}{\partial t}$ is continuous at t_0, it is bounded in a neighborhood of t_0, i.e. there exists

$$\begin{aligned}\frac{S_4(t)-S_4(t_0)}{t-t_0} &= \frac{1}{t-t_0}\int_{b(t_0)}^{b(t)} (f(x,t)-f(x,t_0))\,dx \\ &= \int_{b(t_0)}^{b(t)} \frac{f(x,t)-f(x,t_0)}{t-t_0}\,dx \\ &\leq |b(t)-b(t_0)| \sup_{\xi\text{ß}} \int_{b(t_0)}^{b(t)} \frac{f(x,t)-f(x,t_0)}{t-t_0}\,dx\end{aligned}$$

Example (Variant of Serret's integral).

$$S(t) = \int_0^t \frac{\log(1+tx)}{1+x^2}\,dx$$

Differentiating under the integral, we get:

$$\begin{aligned}S'(t) &= \frac{\log\left(1+t^2\right)}{1+t^2} - 0 + \int_0^t \frac{x}{(1+tx)(1+x^2)}\,dx \\ &= \frac{\log\left(1+t^2\right)}{1+t^2} - 0 - \frac{\log\left(1+t^2\right)}{1+t^2} + \frac{t\arctan(t)}{1+t^2} + \frac{\log\left(1+t^2\right)}{2(1+t^2)} \\ &= \frac{t\arctan(t)}{1+t^2} + \frac{\log\left(1+t^2\right)}{2(1+t^2)}\end{aligned}$$

Integrating the first term by parts yields:

$$\int \frac{t\arctan(t)}{1+t^2}\,dt = \frac{\arctan(t)\log\left(1+t^2\right)}{2} - \int \frac{\log\left(1+t^2\right)}{2(1+t^2)}\,dt$$

Thus the second term cancels out, and so:

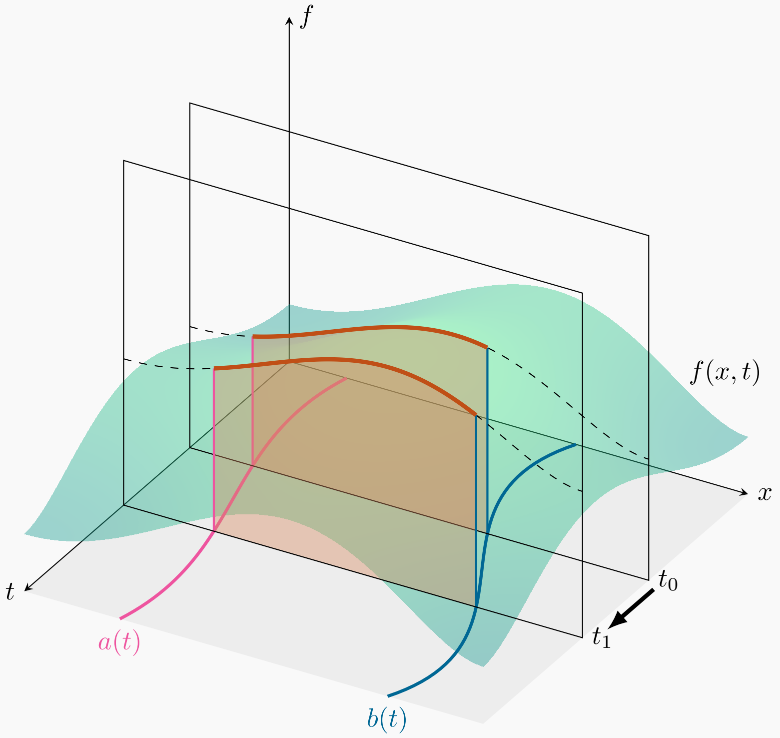
$$S(t) = \frac{\arctan(t)\log\left(1+t^2\right)}{2} + C$$

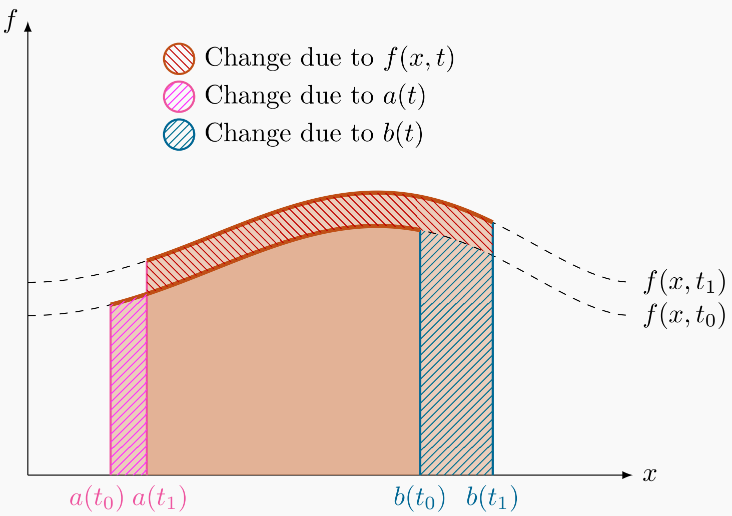
Where C is a constant. As with Serret's integral, the integrand vanishes when $t=0$, so $S(0)=0$ and so $C=0$. Thus:

$$\int_0^t \frac{\log(1+tx)}{1+x^2}\,dx = \frac{\arctan(t)\log\left(1+t^2\right)}{2}$$

The Leibniz Integral Rule (Variable Limits)

$$\frac{d}{dt}\left(\int_{a(t)}^{b(t)} f(x,t)\,dx\right) = b'(t)f(b(t),t) - a'(t)f(a(t),t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x,t)\,dx$$





We now use the Leibniz integral rule to prove the fundamental theorem of algebra:

Theorem 1.3 (Fundamental Theorem of Algebra). Every polynomial $p:\mathbb{C}\to\mathbb{C}$ has at least one root.

Proof. Suppose $p(z)\neq 0$ for all $z\in\mathbb{C}$. Define $I:[0,\infty)\to\mathbb{C}$ as follows:

$$I(r) = \int_0^{2\pi} \frac{1}{p(re^{i\theta})}\,d\theta$$

This is well-defined as $p(z)$ is never zero, so $\frac{1}{p(z)}$ is continuous on $\mathbb{C}$.

We first show that $\lim_{r\to 0^+} I(r) = I(0)$, i.e. I is continuous at 0. Since $\frac{1}{p(z)}$ is continuous at 0, for any $\varepsilon > 0$, there exists $\delta > 0$ such that $\left|\frac{1}{p(z)} - \frac{1}{p(0)}\right| < \frac{\varepsilon}{2\pi}$ for all $z \in \mathbb{C}$, $|z| < \delta$. Thus for all $0 \leq r < \delta$, we have:

$$|I(r) - I(0)| = \left|\int_0^{2\pi} \left(\frac{1}{p(re^{i\theta})} - \frac{1}{p(0)}\right) d\theta\right| \leq \int_0^{2\pi} \left|\frac{1}{p(re^{i\theta})} - \frac{1}{p(0)}\right| d\theta < \int_0^{2\pi} \frac{\varepsilon}{2\pi}\, d\theta = \varepsilon$$

Thus $\lim_{r\to 0^+} I(r) = I(0)$.

We now show that $\lim_{r\to\infty} I(r) = 0$. Suppose the leading (highest-degree) term of $p(z)$ is az^d. Then we have $\lim_{r\to\infty} \frac{|p(re^{i\theta})|}{|re^{i\theta}|^d} = |a|$, so there exists $R > 0$ such that $|p(z)| > \frac{|a|}{2} r^d$ for all $r > R$. Thus for all $r > R$, we have:

$$|I(r)| \leq \int_0^{2\pi} \frac{1}{|p(re^{i\theta})|}\, d\theta \leq \int_0^{2\pi} \frac{2}{|a|r^d}\, d\theta = \frac{4\pi}{|a|r^d}$$

Since the right side tends to zero as $r \to \infty$, we have $\lim_{r\to\infty} I(r) = 0$.

Note that:

$$\frac{\partial}{\partial\theta}\left(\frac{1}{p(re^{i\theta})}\right) = -ire^{i\theta} \frac{p'(re^{i\theta})}{p(re^{i\theta})^2} = ir\frac{\partial}{\partial r}\left(\frac{1}{p(re^{i\theta})}\right)$$

Thus, differentiating under the integral, we get:

$$I'(r) = \int_0^{2\pi} \frac{\partial}{\partial r}\left(\frac{1}{p(re^{i\theta})}\right) d\theta = \int_0^{2\pi} \frac{1}{ir}\frac{\partial}{\partial\theta}\left(\frac{1}{p(re^{i\theta})}\right) d\theta = \frac{1}{ir}\left(\frac{1}{p(re^{i\theta})}\right)\Bigg|_{\theta=0}^{\theta=2\pi} = \frac{1}{ir}\left(\frac{1}{p(r)} - \frac{1}{p(r)}\right) = 0$$

Thus I is constant on $[0, \infty)$. This is a contradiction, as $\lim_{r\to\infty} I(r) = 0$ but $\lim_{r\to 0^+} I(r) = \frac{2\pi}{p(0)} \neq 0$. Thus p must have at least one root, i.e. $p(z) = 0$ for some $z \in \mathbb{C}$. 😎